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Directing light for thermal and power applications in space

Abstract

Solar collectors can provide power for electricity, thermal propulsion, and material processing (e.g., mining asteroids). In one aspect, a rocket propulsion system is configured to produce thrust for a spacecraft and includes: one or more optical elements configured to receive solar energy. The optical elements include: a first window configured to allow energy to enter the rocket propulsion system and form a concentrated energy beam, and a second window positioned to allow the concentrated energy beam to pass to the heat exchanger. The second window is spaced away from the first window to form a pressurized plenum chamber therebetween. The system further includes: a heat exchanger configured to receive the energy and use it to heat and pressurize a propulsion gas, and a rocket nozzle configured to expel the pressurized propulsion gas.

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Background/Summary

INCORPORATION BY REFERENCE TO ANY PRIORITY APPLICATIONS (1) This application is divisional of U.S. Non-Provisional patent application Ser. No. 17/382,727, filed on Jul. 22, 2021, which is based upon and claims the benefit of priority from U.S. Provisional Patent Application No. 63/055,231 filed on Jul. 22, 2020. Moreover, any and all applications for which a foreign or domestic priority claim is identified in the Application Data Sheet as filed with the present application are hereby incorporated by reference under 37 CFR 1.57. The entire contents of each of the above-listed items is hereby incorporated into this document by reference and made a part of this specification for all purposes, for all that each contains.

BACKGROUND

Field

(1) Aspects of this disclosure relate to systems and methods for precisely directing concentrated solar energy from the sun for use aboard spacecraft and for asteroid mining.

SUMMARY OF CERTAIN INVENTIVE ASPECTS

(2) Commercial economic activity in near-earth space and in regions beyond is growing rapidly. In the inner solar system, industrial levels of continuous solar power may be collected by using light-weight concentrating mirrors. The mirrors may be deployed from orbiting spacecraft or may be installed upon relatively fixed installations such as asteroids or low gravity moons. Light-weight mirror structures can collect powerful energy beams, but their reduced dimensional stability (compared to heavier terrestrial designs) may allow the energy beams to wander off target causing damage to supporting structures. It is desirable to have an active control system that can compensate for dimensional changes.

(3) Aspects of this disclosure provide at least two types of robust sensors for use in feedback control systems. The sensors directly sample the full diameter of the energy beams during high power operation. The first method employs imaging cameras that view small amounts of visible energy which is inadvertently scattered from nominally well-polished optical surfaces. The second method places a fine wire grid across the path of an energy beam and measures resistance changes in the wires due to thermal heating. Both methods may involve determining the energy centroid of an energy beam. In the event of beam wander, both methods can provide electrical error signals which can be used to adjust various optical elements to correct for dimensional drift in light-weight structures. Depending on the particular embodiment, one or both of the methods can be applied to generate the electrical error signals used to adjust the optical elements.

(4) Further aspects of this disclosure provide protective shutters which can be closed to permit progressive alignment of high energy beams throughout a spacecraft beam transport system while protecting the spacecraft structure from unintended damage due to misaligned high energy beams.

(5) Still further aspects of this disclosure provide shutters which can be used to interrupt an energy beam for the purpose of selectively heating nearby light-weight surfaces to eliminate moisture and frost buildup.

(6) In one aspect, there is provided a method for protecting spacecraft against unintentional damage from concentrated solar energy-beams by blocking the beams or preventing the concentration of damaging levels of solar energy whenever the beams could potentially damage spacecraft structures.

(7) In another aspect, there is provided a method for protecting spacecraft against unintentional damage from concentrated solar energy beams.

(8) In some embodiments, a plurality of energy-beam sensors monitor continually the directions of solar energy beams with respect to a spacecraft structure without disrupting the said energy beams during normal spacecraft operations. Data from the sensors may be used to provide pointing and tracking information for solar power collection and may also be used to provide warning alerts and shutdown commands whenever energy beams point in potentially damaging directions.

(9) In some embodiments, physical elements may be rapidly interposed into the beam path of a solar energy collection system in order to prevent the formation of a concentrated beam or to absorb or to harmlessly scatter the concentrated energy. The physical elements may include optical shutters, blocking plates, absorbers, diffusers, or optical elements which deflect or defocus solar energy-beams.

(10) In some embodiments, concentrated solar-energy beams may be blocked with a response time that is shorter than the time for thermal damage to occur in spacecraft structures.

(11) In yet another aspect, a method involves using an imaging camera to continually monitor diffusely scattered light from a nominally specular optical reflecting surface in order to derive adjustment signals for the alignment of optical elements in a solar energy collection system.

(12) In some embodiments, an imaging camera on a spacecraft is used to align a large solar collecting mirror directly at the sun without reference to an additional highly accurate sun tracking device.

(13) In some embodiments, an imaging camera on a spacecraft is used to detect misalignment of a concentrated beam of light on an optically reflecting surface in order to prevent a misdirected solar energy beam that could cause unintended damage to parts of the spacecraft structure.

(14) In some embodiments, an imaging camera on a spacecraft is used to detect and measure misalignment of a concentrated beam of light incident upon a closed protective shutter for the purpose of aligning the beam of light before it can be directed to additional optical

elements and thereby reducing the risk of unintended damage to the spacecraft structure.

(15) In some embodiments, an opaque cover plate is placed to temporarily intercept and block a beam of concentrated solar energy, wherein heating of the cover plate and re-radiation of thermal energy from the plate is used to warm nearby surfaces in order to prevent moisture or frost buildup.

(16) In some embodiments, the method involves measuring the position of a concentrated beam of solar energy by means of a grid of fine wires wherein thermal heating of the wires results in electrical resistance changes in the wires.

(17) In some embodiments, a grid of fine wires is disposed across the path of a concentrated beam of solar energy to determine the position of the maximum intensity of the beam.

(18) In still yet another aspect, a rocket propulsion system produces thrust for a spacecraft by directing a concentrated beam of solar energy into a heat exchanger that heats a pressurized propulsion gas which is expelled through a rocket nozzle.

(19) In some embodiments, the concentrated energy beam enters the propulsion system through a transparent window material that functions as a pressure window and may also include curved surfaces to provide a controlled amount of focusing of the energy beam.

(20) In some embodiments, the energy beam passes through a second transparent window that is spaced away from the first window thereby forming a pressurized plenum chamber between the two windows into which a propulsion gas may be injected. The second window is further provided with an array of holes that allow pressurized gas to flow out of the plenum chamber into a further pressure chamber containing the heat exchanger and a rocket nozzle. The array of holes form a turbulence-free protective gas barrier near the output surface of the second window.

(21) In some embodiments, a first propulsion gas is injected into the plenum and a second propulsion gas is injected into the pressure chamber that contains a heat exchanger. The first gas may be formulated to cool the windows and to chemically remove contaminants. The second gas may be a complex mixture of molecules that is derived from asteroid mining processes.

(22) In another aspect, there is provided an apparatus for collecting solar energy and simultaneously protecting against damage from a resulting energy beam, the apparatus comprising: a solar energy collection system comprising at least one concentrator and a target configured to use, store, or convert the solar energy, the collection system configured to cause solar energy to focus on the target; at least one sensor configured to detect misalignment of the concentrator by determining that some or all of the collected solar energy is offset from the target; and a safety system configured to redirect the energy or interpose a safety structure for shielding other non-target systems from receiving too much solar energy from the collection system.

(23) In some embodiments, the safety structure comprises one or more of the following: an optical shutter, a blocking plate, an absorber, a diffuser, and/or an optical element, and wherein the physical element is further configured to deflect and/or defocus the collected solar energy.

(24) In some embodiments, the safety system is further configured to interpose the safety structure with a response time that is shorter than the time for thermal damage to occur in the non-target systems.

(25) In some embodiments, the safety structure is further configured to receive the collected sunlight and radiate thermal energy in order to prevent moisture or ice build up on other portions of the solar power collection system.

(26) In some embodiments, the at least one concentrator is configured to focus the solar energy, the solar energy collection system further comprises at least one reflector configured to reflect the solar energy focused by the at least one concentrator, and the at least one sensor comprises at least one camera having a field of view directed at the at least one reflector.

(27) In some embodiments, the apparatus further comprises: one or more processors configured to: receive image data from the one or more cameras, detect a location of a diffuse glow with respect to the at least one reflector, and determine whether the focused solar energy is misaligned based on the detected location of the diffuse glow.

(28) In some embodiments, the solar energy collection system further comprises: one or more actuators configured to adjust the position and/or rotation of the at least one reflector based on the determination of whether the focused solar energy is misaligned.

(29) In some embodiments, the at least one concentrator is configured to focus the solar energy, the solar energy collection system further comprises at least one reflector configured to reflect the solar energy focused by the at least one concentrator, and the one or more sensors comprise a grid formed of a plurality of wires located adjacent to one of the at least one reflective elements.

(30) In some embodiments, the apparatus further comprises: a circuit configured to measure electrical resistances of each of the plurality of wires, and one or more processors configured to: receive the measured electrical resistances from the circuit, and determine whether the focused solar energy is misaligned based on the measured electrical resistances.

(31) In some embodiments, the solar power collection system further comprises: one or more actuators configured to adjust the position and/or rotation of the at least one reflector based on the determination of whether the focused solar energy is misaligned.

(32) In yet another aspect, there is provided a spacecraft, comprising: an asteroid capture system configured to capture an asteroid; a solar power collection system configured to: collect solar energy from the sun, generate an energy beam from the collected sunlight, and redirect the energy beam into the captured asteroid, wherein the solar power collection system comprises one or more sensors configured to detect misalignment of the energy beam, the solar power collection system further configured to redirect the energy beam based on output of the one or more sensors.

(33) In some embodiments, the solar power collection system further comprises: a physical element configured to selectively prevent the solar power collection system from forming the energy beam in response to the one or more sensors detecting misalignment of the energy beam.

(34) In some embodiments, the physical element comprises one or more of the following: an optical shutter, a blocking plate, an absorber, a diffuser, and/or an optical element, and wherein the physical element is further configured to deflect and/or defocus the collected solar energy.

(35) In some embodiments, the physical element is further configured to prevent the solar power collection system from forming the energy beam with a response time that is shorter than the time for thermal damage to occur in spacecraft structures.

(36) In some embodiments, the physical element is further configured to receive the collected sunlight and radiate thermal energy in order to prevent moisture or ice build up on other portions of the solar power collection system.

(37) In some embodiments, the solar power collection system further comprises: a lenticular structure having a curved reflective surface configured to collect the solar energy and focus the collected solar energy into the energy beam, and at least one reflecting element configured to redirect the energy beam.

(38) In some embodiments, the one or more sensors comprise one or more cameras, the one or more cameras each have a field of view

directed at one of the at least one reflecting element.

(39) In some embodiments, the spacecraft further comprises: one or more processors configured to: receive image data from the one or more cameras, detect a location of a diffuse glow with respect to the one of the reflective elements, and determine whether the energy beam is misaligned based on the detected location of the diffuse glow.

(40) In some embodiments, the solar power collection system further comprises: one or more actuators configured to adjust the position and/or rotation of the at least one reflecting element based on the determination of whether the energy beam is misaligned.

(41) In some embodiments, the one or more sensors comprise a grid formed of a plurality of wires located adjacent to one of the at least one reflective elements.

(42) In some embodiments, the spacecraft further comprises: a circuit configured to measure electrical resistances of each of the plurality of wires, and one or more processors configured to: receive the measured electrical resistances from the circuit, and determine whether the energy beam is misaligned based on the measured electrical resistances.

(43) In some embodiments, the solar power collection system further comprises: one or more actuators configured to adjust the position and/or rotation of the at least one reflecting element based on the determination of whether the energy beam is misaligned.

(44) In some embodiments, further comprising: a rocket propulsion system configured to provide rocket thrust, wherein the solar power collection system is further configured to redirect the energy beam into the rocket propulsion system.

(45) In some embodiments, the rocket propulsion system comprises: a rocket nozzle, and a heat exchanger configured to heat a pressurized propulsion gas and expel the heated pressurized propulsion through the rocket nozzle.

(46) In some embodiments, the rocket propulsion system further comprises: a body enclosing the heat exchanger, and a first transparent window configured to provide a pressure seal against the body and provide a controlled amount of focusing of the energy beam to the heat exchanger.

(47) In some embodiments, the rocket propulsion system further comprises: a second transparent window arranged within the body and spaced away from the first transparent window to form a pressurized plenum chamber between the first and second transparent windows, wherein the plenum chamber is configured to receive a first propulsion gas injected therein, wherein the second window has an array of holes formed therein configured to allow the first propulsion gas to flow out of the plenum chamber into a pressure chamber formed in the body and containing the heat exchanger, and wherein the array of holes form a substantially turbulence-free protective gas barrier near an output surface of the second transparent window.

(48) In some embodiments, the pressure chamber is configured to receive a second propulsion gas injected therein, the first gas is configured to cool and remove contaminants from the first and second transparent windows, and the second gas comprises a mixture of molecules derived from asteroid mining processes.

(49) In still yet another aspect, a method for protecting a spacecraft against damage from an energy beam, comprising: collecting solar energy from the sun using a solar power collection system of the spacecraft; generating an energy beam from the collected sunlight using the solar power collection system; redirecting the energy beam into an asteroid captured by an asteroid capture system of the spacecraft; detecting misalignment of the energy beam using one or more sensors; and redirecting the energy beam using the solar power collection system based on output of the one or more sensors.

(50) In some embodiments, the method further comprises: configured to selectively preventing the solar power collection system from forming the energy beam using a physical element in response to the one or more sensors detecting misalignment of the energy beam.

(51) In some embodiments, the physical element comprises one or more of the following: an optical shutter, a blocking plate, an absorber, a diffuser, and/or an optical element, and wherein the physical element is further configured to deflect and/or defocus the collected solar energy.

(52) In some embodiments, the method further comprises: preventing the solar power collection system from forming the energy beam using the physical element with a response time that is shorter than the time for thermal damage to occur in spacecraft structures.

(53) In some embodiments, the method further comprises: receiving the collected sunlight and radiate thermal energy using the physical element in order to prevent moisture or ice build up on other portions of the solar power collection system.

(54) In some embodiments, the method further comprises: collecting and focusing the solar energy using a lenticular structure having a curved reflective surface into the energy beam, and redirecting the energy beam using at least one reflecting element.

(55) In some embodiments, the one or more sensors comprise one or more cameras, the one or more cameras each have a field of view directed at one of the at least one reflecting element.

(56) In some embodiments, the method further comprises: receiving, at one or more processors, image data from the one or more cameras; detecting, at the one or more processors, a location of a diffuse glow with respect to the one of the reflective elements; and determining, using the one or more processors, whether the energy beam is misaligned based on the detected location of the diffuse glow.

(57) In some embodiments, the method further comprises: adjusting the position and/or rotation of the at least one reflecting element using one or more actuators based on the determination of whether the energy beam is misaligned.

(58) In some embodiments, the one or more sensors comprise a grid formed of a plurality of wires located adjacent to one of the at least one reflective elements.

(59) In some embodiments, the method further comprises: measuring electrical resistances of each of the plurality of wires using a circuit; receiving, at one or more processors, the measured electrical resistances from the circuit; and determining, at the one or more processors, whether the energy beam is misaligned based on the measured electrical resistances.

(60) In some embodiments, the method further comprises: adjusting the position and/or rotation of the at least one reflecting element using one or more actuators based on the determination of whether the energy beam is misaligned.

(61) In some embodiments, the method further comprises: providing rocket thrust using a rocket propulsion system; and redirecting the energy beam into the rocket propulsion system.

(62) In some embodiments, the method further comprises: heating a pressurized propulsion gas using a heat exchanger; and expelling the heated pressurized propulsion through a rocket nozzle.

(63) In some embodiments, the rocket propulsion system further comprises: a body enclosing the heat exchanger, and a first transparent window configured to provide a pressure seal against the body and provide a controlled amount of focusing of the energy beam to the heat exchanger.

(64) In some embodiments, the rocket propulsion system further comprises: a second transparent window arranged within the body and

spaced away from the first transparent window to form a pressurized plenum chamber between the first and second transparent windows, wherein the plenum chamber is configured to receive a first propulsion gas injected therein, wherein the second window has an array of holes formed therein configured to allow the first propulsion gas to flow out of the plenum chamber into a pressure chamber formed in the body and containing the heat exchanger, and wherein the array of holes form a substantially turbulence-free protective gas barrier near an output surface of the second transparent window.

(65) In some embodiments, the method further comprises: the pressure chamber is configured to receive a second propulsion gas injected therein, the first gas is configured to cool and remove contaminants from the first and second transparent windows, and the second gas comprises a mixture of molecules derived from asteroid mining processes.

(66) These and other features and advantages of the present disclosure will be apparent to those skilled in the art from the following detailed description, taken together with the accompanying drawings, in which like reference numerals refer to like parts.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a schematic illustration of an embodiment of a spacecraft installed within a launch vehicle, wherein the various light-weight structures of the spacecraft are folded into compact configurations in accordance with aspects of this disclosure.

(2) FIG. 2 is a schematic illustration of the orbiting spacecraft, wherein lightweight structures have been unfolded and deployed to operational configurations in accordance with aspects of this disclosure.

(3) FIG. 3 is a schematic illustration of the collection and concentration of solar energy by light-weight reflecting structures in accordance with aspects of this disclosure.

(4) FIG. 4 illustrates a camera used to image the position of an energy beam on a polished optical surface in accordance with aspects of this disclosure.

(5) FIG. 5 is a photograph of scattering of high intensity light from a polished surface in accordance with aspects of this disclosure.

(6) FIG. 6 illustrates a protective hinge and cover plate that can interrupt the formation of an energy beam in accordance with aspects of this disclosure.

(7) FIG. 7 illustrates a wire-grid sensor for measuring the centroid of a high energy beam of solar energy in accordance with aspects of this disclosure.

(8) FIG. 8 illustrates an embodiment in which a turning mirror is configured to be rotated to direct the energy beam toward the rocket propulsion system in accordance with aspects of this disclosure.

(9) FIG. 9 illustrates internal structures of the rocket propulsion system in accordance with aspects of this disclosure.

DETAILED DESCRIPTION

(10) When launching spacecraft by rockets from the surface of the Earth, the spacecraft are typically carried within the aerodynamic nose cone of the launch vehicle. It is desirable to reducing launch costs by minimizing both the total mass of the spacecraft and its physical size. It is therefore useful to employ compact and light-weight structures wherever possible. As used herein, light-weight structures may generally refer to structures having a sufficiently low weight such that the structures can be launched from the Earth to escape from the gravitational influence of Earth using one or more rockets.

(11) Aspects of this disclosure relate to systems and techniques for collecting solar energy and protecting against damage resulting from an energy beam formed from the collected solar energy. Although examples of such systems and techniques are described in connection with an asteroid mining spacecraft, aspects of this disclosure are not limited thereto. For example, the systems and techniques described herein can also be applied to solar collection systems that are installed temporarily or permanently on an asteroid or a moon. As described herein, there may be situations in which focused solar energy may have sufficient power to damage components which do not form part of the optical path for the solar energy, for example, due to misalignment of one or more components in the system. Thus, aspects of this disclosure can provide systems and techniques for preventing such damage from occurring in many different solar energy collection systems.

(12) FIG. 1 illustrates an embodiment of an asteroid mining spacecraft, which can be positioned atop a launch rocket 5 (also referred to as a booster rocket) in accordance with aspects of this disclosure. The main body of the spacecraft 1, also called the bus, contains the fuel, electrical components, and communication equipment for a typical mission. The spacecraft 1 is further provided with one or more rocket propulsion systems 2 configured to provide maneuvering capability to allow the spacecraft 1 to reach and capture passing asteroids. The spacecraft 1 also contains an asteroid capture system 3 and a solar power collection system 4, both of which are shown in a folded and stowed configuration in FIG. 1. The booster rocket 5 is shown schematically by dashed lines.

(13) FIG. 2 depicts an embodiment the asteroid mining spacecraft 1 in orbital flight after it has separated from the booster rocket 5 in accordance with aspects of this disclosure. The asteroid capture system 3 is shown in a deployed configuration. In certain embodiments, the asteroid capture system 3 may include a plurality of multiply-jointed rigid members or may include a plurality of hollow tubular polymer membranes which have been inflated by internal gas pressure. However, aspects of this disclosure are not limited thereto and in other embodiments the asteroid capture system 3 can include other structure member(s) configured to be deployed in order to capture and contain asteroids.

(14) With continued reference to FIG. 2, the solar power collection system 4 is shown after it has been unfolded and deployed from its compact stowed configuration. In certain implementations, the solar power collection system 4 comprises a plurality of linear support elements 7, 8, and 9 which may be embodied as tubular polymer membranes that have been inflated and held rigid by internal pressurizing gas. The solar power collection system 4 further comprises a hollow lens-shaped or inflated lenticular structure 10 which provides a transparent upper surface 11 and a curved reflecting lower surface 12. The lower surface 12 is configured to serve as a curved mirror to collect and focus incoming solar radiation. The solar power collection system 4 further comprises rigid reflecting elements 13 and 14 and may further include additional optical elements (not shown) which can be used to direct a focused energy beam.

(15) FIG. 3 illustrates an embodiment of the solar power collection system 4 configured to collect solar energy and focusing the solar energy into a powerful energy beam. When the collecting curved mirror 12 is pointed directly at the Sun, a portion of the incoming solar radiation 15 reflects from the curved surface 12 and converges toward a secondary mirror 13.

(16) With an unobstructed view of the Sun, the collection of solar energy using the configuration illustrated in FIG. 3 can collect very large power beams. For example, in an embodiment in which the mirror 12 has a collecting area of about 1 square meter, the continuous

power collected can exceed one kilowatt of thermal energy for spacecraft at Earth's orbital distance from the Sun. When this energy is focused to a spot of several centimeters in diameter, the beam can have sufficient power density to drill a hole into concrete or into a captured asteroid. In certain implementations, collecting mirrors **12** having an area larger than about 1 square meter and multiple mirror configurations can collect proportionally larger power beams.

(17) The additional rigid reflecting elements **13** and **14** are configured to adjust the focus and direction of the concentrated energy beam **16**. Energy beam **16** can be directed into the spacecraft **1** via the rigid reflecting elements **13** and **14**. The solar power collecting system **4** can further include one or more additional optical elements **17** configured to direct the beam **16** to pass through the spacecraft **1** to the asteroid capture system **3** and to impinge upon a captured asteroid **18**. The concentrated solar energy can be used to drill holes into the asteroid, to melt and vaporize volatile constituents of the asteroid material, and to cause spalling and excavation of the asteroid surface. Thus, as used herein, the high power energy beam **16** may refer to an energy beam having sufficient power to drill a hole into the captured asteroid, to melt and vaporize volatile constituents of the asteroid material, and/or to cause spalling and excavation of the asteroid surface.

(18) It can be seen from the above description that it is desirable to carefully focus a powerful energy beam **16** and direct the energy beam **16** to avoid hitting and damaging parts of the spacecraft **1** structure. That is, the energy beam **16** may have sufficient power to damage the spacecraft **1** if a portion of the beam **16** is inadvertently misdirected and strikes components of the spacecraft **1** which do not form part of the solar power collecting system **4**. There are several conditions that may affect a dynamic misalignment of the beam **16**. A non-exhaustive list of these conditions includes, for example, errors in pointing reflecting mirror **12** directly at the sun, mechanical deformations of light-weight structures **3** and **7-12** due to spacecraft **1** maneuvering forces, and/or slowly changing dimensions of the structures **3** and **7-12** due to material fatigue and/or due to pressurization loss caused by micro meteor impacts.

(19) Therefore, it is desirable to provide techniques for continually monitoring the position and focus of energy beam **16**. It is also desirable to incorporate shutters which can safely block or prevent the formation of an energy beam **16** during solar alignment maneuvers. It is also desirable to quickly block the energy beam **16** should there be unexpected upsets in any of the spacecraft control systems, a so called "dead man" automatic safety shutdown system.

(20) In various embodiments of this disclosure, the spacecraft **1** can include one or more sensors configured to generate signals indicative of the alignment, misalignment, and/or degree of alignment of the energy beam **16**. For example, solar power collecting system **4** can use the signals generated by the sensor(s) to realign one or more of the optical components that redirect the energy beam **16** in order to ensure that the energy beam **16** is not directed at other portions of the spacecraft **1**. The one or more sensors may be configured to provide data for pointing and tracking information for solar power collection and may also provide warning alerts and shutdown commands whenever the energy beam **16** points in potentially damaging direction.

(21) The one or more sensors can be implemented by various different embodiments. For example, the one or more sensors can use or be embodied as video camera(s) **19** as described in connection with FIG. **4** below. In some implementations, the sensor(s) can use or be embodied as a grid of fine wires **26** as described in connection with FIG. **7** below.

(22) Referring to FIG. **4**, the spacecraft **1** includes a video camera **19** which may be placed in a convenient location upon the spacecraft body **1** or upon a supporting structure **7**, **8**, or **9**. The camera **19** has a field-of-view **20**, shown by broken lines in FIG. **4**, which is directed to view the surface of reflecting element **13** while being simultaneously well away from the energy beam **16**. If reflecting element **13** were a perfect mirror, none of the collected light in energy beam **16** would be scattered toward the camera **19**. As a practical matter, virtually all polished mirrors diffusely scatter at least a small percentage of incident light into non-specular directions. Accordingly, the image seen by the camera **19** will be a diffuse glow located upon a region of the surface of element **13**. If curved reflector **12** is perfectly aligned toward the incoming rays of sunlight **15**, then the camera **19** image of the diffuse glow will be centered upon element **13**. However, in the event that there is a small pointing error in the alignment of reflector **12**, then the image glow will be displaced from the center of element **13**. A displacement of the glow is an early warning measure of potentially damaging misalignment of the energy beam **16**. Image processing software can readily detect the displacement of the glow image and provide signals to adjust the pointing direction of reflector **12**. For example, in certain embodiments, the camera **19** may include one or more processors configured to run the image processing software in order to detect displacement of the glow image. In other embodiments, the spacecraft **1** can include one or more processors separate from the camera **19** configured to run the image processing software.

(23) In a like manner, additional imaging cameras (not illustrated) may be placed to detect misalignments of the energy beam **16** on other reflecting surfaces along the path of the energy beam **16**. Misalignment measurements at each imaged reflecting element can be used to correct the pointing accuracy of the preceding element in the optical path. Misalignment measurements by a plurality of imaging cameras can provide continuous adjustment of the high energy beam **16** during operation without requiring a system shutdown for calibration. In certain implementations, each of the plurality of cameras may include its own set of one or more processors configured to run image processing software to detect displacement of the respective glow images. In other implementations, each of the plurality of cameras may provide image data to one or more processors separate from the cameras which run the image processing software for each of the received video feeds.

(24) FIG. **5** is a photograph of the diffuse glow image upon a polished mirror in accordance with aspects of this disclosure. In this image, concentrated sunlight from a lens **21** is reflected from a concave mirror surface **22**. The camera taking the image has been positioned to avoid the strong specular reflection. The diffuse glow can be clearly seen as a spot of light **23** that is not centered on the polished mirror.

(25) Referring to FIG. **6** in a second embodiment, the spacecraft **1** includes a hinge **24** and cover plate **25** affixed to cover the front surface of reflecting element **13**. In this configuration, converging rays from curved reflector **12** are prevented from reflecting and combining to form energy beam **16**. Depending on the surface reflectance of cover plate **25**, the incident light will be partially absorbed and partially diffusely reflected. As in the first embodiment of FIG. **4**, the camera **19** will image a diffuse glow upon the surface of cover plate **23**. The image position of the glow spot on the surface of the cover plate indicates the accuracy of alignment of curved surface **12** to the incoming solar energy **15** without requiring a reflection from element **13**. In this manner, the curved surface **12** can be accurately aligned to the direction of the sun before a powerful energy beam **16** is directed further into the spacecraft **1**. Furthermore, the curved surface **12** may be accurately aligned in the direction of the sun without need for a separate sun tracker or sun observing device.

(26) With continued reference to FIG. **6** with the cover plate **25** in the blocking position as shown, a significant amount of solar energy **15** can be absorbed by the cover plate **25** thereby causing its temperature to rise. As the plate **25** heats, it will radiate long wavelength thermal energy in all directions. Nearby structures, such as the lenticular structure **10** with the transparent upper surface **11** will intercept

a portion of the radiated thermal energy and also begin to warm. In this manner, lenticular structure **10** and other surfaces can be warmed as needed to prevent moisture or ice build that could potentially degrade the optical performance of these components. Since moisture may be produced during asteroid mining operations, it is desirable to heat optical surfaces as needed.

(27) In some implementations, the cover plate **25** may be replaced or supplemented with another type of physical blocking element. For example, the physical blocking element may include one or more of the following: optical shutter(s), blocking plate(s), absorber(s), diffuser(s), and/or optical element(s) configured to deflect or defocus the energy beam **16**.

(28) FIG. **7** illustrates a third embodiment of the solar power collection system **4** in accordance with aspects of this disclosure. As shown in FIG. **7**, the solar power collection system **4** includes a grid of fine wires **26** placed near the reflecting surface of reflecting element **13**. The wires **26** can be fabricated from materials which can tolerate high temperatures, such as tungsten or stainless steel. The wires **26** may have small diameters such as about 0.001 inch. The fine wires **26** intercept a small percentage of the converging solar radiation that is incident upon the reflecting element **13** and begin to warm. As the wires **26** are heated, their electrical resistance will increase. The wires closest to the center of the energy beam **16** will exhibit the largest resistance increases. The solar power collection system **4** can include one or more circuit configured to measure the resistance difference between adjacent wires **26**, which provides information about directional misalignment of the energy beam **16**. Very small resistance differences may be accurately measured by a common electrical circuit known as a Wheatstone Bridge. One or more processors may be configured to determine whether the energy beam **16** is misaligned based on the measured electrical resistances and provide control signals for repositioning and/or rotating the optical elements **10**, **13**, **16**, **17** to correct the misalignment of the energy beam **16**.

(29) The solar power collection system **4** can include one or more actuators (not illustrated) configured to reposition and/or rotate one or more of the optical elements **10**, **13**, **16**, **17** that generate and/or redirect the energy beam **16**. Thus, the solar power collection system **4** can use the outputs received from the camera(s) **19** and/or the wires **26** as an input to control the alignment of the optical elements **10**, **13**, **16**, **17** to ensure that the beam **16** is not directed onto any unintended components of the spacecraft **1**, and thus, does not damage any components of the spacecraft **1**.

(30) FIG. **8** illustrates an embodiment in which a turning mirror **17** is configured to be rotated to direct the energy beam **16** toward the rocket propulsion system **2** in accordance with aspects of this disclosure. FIG. **9** illustrates internal structures of the rocket propulsion system **2** in accordance with aspects of this disclosure.

(31) With reference to FIGS. **8** and **9**, optical element **17** is embodied as a turning mirror **17** which can be rotated to selectively direct the energy beam **16** toward the rocket propulsion system **2** instead of towards the asteroid capture system **3**. For example, the turning mirror **17** may be coupled to an actuator (not illustrated) configured to rotate the turning mirror **17** between a first orientation (e.g., as shown in FIG. **3**) and a second orientation (e.g., as shown in FIG. **8**).

(32) The energy beam **16**, represented in FIG. **9** within boundary rays, enters the rocket propulsion system **2** through two transparent windows **27** and **28**. A first window **27** forms a pressure seal against components of the rocket propulsion system body **30** and **30a**. The window **27** may have flat or curved surfaces forming a focusing lens. To reduce undesired optical reflections, one or both surfaces of the first window **27** may be coated with anti-reflection coatings.

(33) The first window focuses and converges the energy beam **16** into a focused converging energy beam **16a**, represented within boundary rays. The focused converging energy beam **16a** passes through a second transparent window **28** and then into a heat exchanger **32**.

(34) The rocket propulsion system **2** is configured to heat a pressurized propellant gas to a high temperature and then expel the gas through a converging-diverging rocket nozzle **31** in order to produce useful rocket thrust. It is a general physical principle that most suitable propellant gasses are substantially transparent to solar energy. Thus, the gas will not efficiently absorb the wavelengths present in solar light.

(35) The solid bulk material of heat exchanger **32** is designed to be an efficient absorber of solar energy, e.g., provided by the energy beam **16a**. The incoming energy beam **16a** heats the heat exchanger **32** to a high temperature, typically on the order of 2000 degrees Celsius. The heat exchanger **32** is also porous to allow the flow of propellant gasses through the heat exchanger **32**. The heat exchanger **32** also contains a large internal surface area which, by heat conduction, efficiently transfers heat energy from the bulk material of the heat exchanger **32** to the propellant gas.

(36) Highly concentrated solar energy in a small volume can produce temperatures far higher than the melting point of all known solid materials. Therefore, the focusing properties of first window **27** may be chosen to limit the concentration of solar energy to temperatures within the safe operating limits of the heat exchanger **32** and the surrounding structures **30a**.

(37) The second window **28** is spaced away from first window **27** by a predetermined distance. In some implementations, the distance between the first and second windows **27** and **28** may be on the order of 1 centimeter, however, in other implementations, the distance may be greater or less than 1 centimeter. In one illustrative embodiment, the distance may be 20 centimeters. In certain embodiments, the volume **40** contained between the two windows **27** and **28** forms a plenum configured to hold a pressurized propulsion gas. A first propulsion gas **41** can be injected into the plenum **40** through a first injector tube **42**. In some embodiments, the second window **28** can include a plurality of small through-holes **29** which permit the first propulsion gas **41** to uniformly flow through window **28** into the volume of thrust chamber **50** in a two-dimensional pattern of linear jets.

(38) One purpose of the first propulsion gas **41** is to provide cooling for both windows **27** and **28**. The cooling gas reduces undesirable thermal stresses on both windows. Cooling reduces the probability of window fractures. Furthermore, cooling permits anti-reflection coatings to be applied to both windows, thereby increasing optical efficiency. Antireflection coatings generally have temperature limits that are lower than the temperature capabilities of solid materials and cannot be applied to excessively hot surfaces.

(39) Similarly, a second propulsion gas **43** can be injected through a second injector tube **44** into the thrust chamber **50**. Both of the gases **41** and **43** can be driven by their supply pressures to flow through the heat exchanger **32** and into the rocket nozzle **31**. The temperature of the gases **41** and **43** and their thermal energy content (enthalpy) can be greatly increased by passage through the heat exchanger **32** (e.g., which provides heat from the energy beam **16a** to the gases **41** and **43**). The rocket nozzle **31** efficiently converts the random thermal motion and the relatively slow flow-velocity of hot gas molecules into linear motion in a high-speed exhaust plume, thereby producing rocket thrust.

(40) The use of the two windows **27** and **28** and two injection gasses **41** and **43** provides a number of advantages over other thruster systems. One advantage is that many different types of gasses can be used to produce thrust in the rocket propulsion system **2**. Examples of gases that can be used in the rocket propulsion system **2** include but are not limited to: H.sub.2, H.sub.2O, HF, Ar, Cl.sub.2, Br,

CO.sub.2, N.sub.2, NH.sub.3, CH.sub.4, mixtures of these gases, and a variety of volatile hydrocarbon molecules. Such gases can be derived directly from asteroid mining operations.

(41) Another advantage is that many gas mixtures, when subjected to intense photochemical and thermochemical processes, may dissociate and recombine to produce opaque compounds. It is undesirable to allow opaque compounds to settle on the transparent window surfaces where they can block the energy beam **16**. By injecting a relatively cool first propulsion gas **41** through the multiple holes **29** in the second window **28**, a gas barrier can be formed near the surface of the second window **28**. The outflow of the gas **41** can prevent turbulent gas flow from the propulsion gases **41** and **43** and various contaminants due to heat exchanger **32** reactions from reaching and contaminating the second window **28**.

(42) When exposed to extreme temperatures, some gases including CH.sub.4 are known to deposit carbon compounds on nearby surfaces. This process is known as coking. According to aspects of the present disclosure, coking deposits may be removed by periodically injecting cleaning gasses such as H.sub.2, HF, or Cl.sub.2 singly or in combination through injection ports **42** and/or **44**, providing yet another advantage to aspects of this disclosure.

CONCLUSION

(43) Unless the context clearly requires otherwise, throughout the description and the claims, the words “comprise,” “comprising,” and the like are to be construed in an inclusive sense, as opposed to an exclusive or exhaustive sense; that is to say, in the sense of “including, but not limited to.” The word “coupled”, as generally used herein, refers to two or more elements that may be either directly connected, or connected by way of one or more intermediate elements. Likewise, the word “connected”, as generally used herein, refers to two or more elements that may be either directly connected, or connected by way of one or more intermediate elements. Additionally, the words “herein,” “above,” “below,” and words of similar import, when used in this application, shall refer to this application as a whole and not to any particular portions of this application. Where the context permits, words in the above Detailed Description using the singular or plural number may also include the plural or singular number respectively. The word “or” in reference to a list of two or more items, that word covers all of the following interpretations of the word: any of the items in the list, all of the items in the list, and any combination of the items in the list.

(44) Moreover, conditional language used herein, such as, among others, “can,” “could,” “might,” “can,” “e.g.,” “for example,” “such as” and the like, unless specifically stated otherwise, or otherwise understood within the context as used, is generally intended to convey that certain embodiments include, while other embodiments do not include, certain features, elements and/or states. Thus, such conditional language is not generally intended to imply that features, elements and/or states are in any way required for one or more embodiments or that one or more embodiments necessarily include logic for deciding, with or without author input or prompting, whether these features, elements and/or states are included or are to be performed in any particular embodiment.

(45) The above detailed description is not intended to be exhaustive or to limit the invention to the precise form disclosed above. While specific embodiments of, and examples are described above for illustrative purposes, various equivalent modifications are possible within the scope of the disclosed invention(s), as those skilled in the relevant art will recognize. For example, while processes or blocks are presented in a given order, alternative embodiments may perform routines having steps, or employ systems having blocks, in a different order, and some processes or blocks may be deleted, moved, added, subdivided, combined, and/or modified. Each of these processes or blocks may be implemented in a variety of different ways. Also, while processes or blocks are at times shown as being performed in series, these processes or blocks may instead be performed in parallel, or may be performed at different times.

(46) The teachings provided herein can be applied to other systems, not necessarily the system described above. The elements and acts of the various embodiments described above can be combined to provide further embodiments.

(47) While certain embodiments have been described, these embodiments have been presented by way of example only, and are not intended to limit the scope of the disclosure. Indeed, the novel methods and systems described herein may be embodied in a variety of other forms; furthermore, various omissions, substitutions and changes in the form of the methods and systems described herein may be made without departing from the spirit of the disclosure. The accompanying claims and their equivalents are intended to cover such forms or modifications as would fall within the scope and spirit of the disclosure.

Claims

1. A spacecraft, comprising: an asteroid capture system configured to capture and contain an asteroid; and a solar power collection system configured to: collect solar energy from the sun, generate a solar beam from the collected solar energy, and redirect the solar beam into the captured asteroid, wherein the solar power collection system comprises one or more sensors configured to detect misalignment of the solar beam, the solar power collection system further configured to redirect the solar beam based on output of the one or more sensors.
2. The spacecraft of claim 1, wherein the solar power collection system further comprises: a physical element configured to selectively prevent the solar power collection system from forming the solar beam in response to the one or more sensors detecting misalignment of the solar beam.
3. The spacecraft of claim 2, wherein the physical element comprises one or more of the following: an optical shutter, a blocking plate, an absorber, a diffuser, and/or an optical element, and wherein the physical element is further configured to deflect and/or defocus the collected solar energy.
4. The spacecraft of claim 2, wherein the physical element is further configured to prevent the solar power collection system from forming the solar beam with a response time that is shorter than the time for thermal damage to occur in spacecraft structures.
5. The spacecraft of claim 2, wherein the physical element is further configured to receive the collected solar energy and radiate thermal energy in order to prevent moisture or ice build up on other portions of the solar power collection system.
6. The spacecraft of claim 1, wherein the solar power collection system further comprises: a lenticular structure having a curved reflective surface configured to collect the solar energy and focus the collected solar energy into the solar beam, and at least one reflecting element configured to redirect the solar beam.
7. The spacecraft of claim 6, wherein: the one or more sensors comprise one or more cameras, and the one or more cameras each have a field of view directed at one of the at least one reflecting element.
8. The spacecraft of claim 7, further comprising: one or more processors configured to: receive image data from the one or more cameras, detect a location of a diffuse glow with respect to the one of the reflective elements, and determine whether the solar beam is

misaligned based on the detected location of the diffuse glow; and one or more actuators configured to adjust the position and/or rotation of the at least one reflecting element based on the determination of whether the solar beam is misaligned.

9. The spacecraft of claim 6, wherein the one or more sensors comprise a grid formed of a plurality of wires located adjacent to one of the at least one reflective elements.

10. The spacecraft of claim 9, further comprising: a circuit configured to measure electrical resistances of each of the plurality of wires, and one or more processors configured to: receive the measured electrical resistances from the circuit, and determine whether the solar beam is misaligned based on the measured electrical resistances, wherein the solar power collection system further comprises one or more actuators configured to adjust the position and/or rotation of the at least one reflecting element based on the determination of whether the solar beam is misaligned.

11. The spacecraft of claim 1, further comprising: a rocket propulsion system configured to provide rocket thrust, wherein the solar power collection system is further configured to redirect the solar beam into the rocket propulsion system.

12. The spacecraft of claim 11, wherein the rocket propulsion system comprises: a rocket nozzle, a heat exchanger configured to heat a pressurized propulsion gas and expel the heated pressurized propulsion through the rocket nozzle, a body enclosing the heat exchanger, and a first transparent window configured to provide a pressure seal against the body and provide a controlled amount of focusing of the solar beam to the heat exchanger.

13. The spacecraft of claim 12, wherein the rocket propulsion system further comprises: a second transparent window arranged within the body and spaced away from the first transparent window to form a pressurized plenum chamber between the first and second transparent windows, wherein the plenum chamber is configured to receive a first propulsion gas injected therein, wherein the second window has an array of holes formed therein configured to allow the first propulsion gas to flow out of the plenum chamber into a pressure chamber formed in the body and containing the heat exchanger, and wherein the array of holes form a substantially turbulence-free protective gas barrier near an output surface of the second transparent window.

14. The spacecraft of claim 13, wherein: the pressure chamber is configured to receive a second propulsion gas injected therein, the first gas is configured to cool and remove contaminants from the first and second transparent windows, and the second gas comprises a mixture of molecules derived from asteroid mining processes.

15. The spacecraft of claim 1, wherein the asteroid capture system comprises a plurality of multiply-jointed rigid members configured to capture and contain the asteroid.

16. The spacecraft of claim 1, wherein the asteroid capture system comprises a plurality of hollow tubular polymer membranes configured to be inflated by internal gas pressure, the hollow tubular polymer membranes configured to capture and contain the asteroid.

17. The spacecraft of claim 1, wherein the solar power collection system is configured to redirect the solar beam into the captured asteroid to drill holes into the captured asteroid, to melt and vaporize volatile constituents of the captured asteroid, and to cause spalling and excavation of the captured asteroid.

18. A method for mining an asteroid, comprising: capturing and containing an asteroid using an asteroid capture system of a spacecraft; collecting solar energy from the sun using a solar power collection system of the spacecraft; generating an solar beam from the collected solar energy using the solar power collection system; redirecting the solar beam into the captured asteroid; detecting misalignment of the solar beam using one or more sensors; and redirecting the solar beam using the solar power collection system based on output of the one or more sensors.

19. The method of claim 18, further comprising: selectively preventing the solar power collection system from forming the solar beam using a physical element in response to the one or more sensors detecting misalignment of the solar beam.

20. The method of claim 19, wherein the physical element comprises one or more of the following: an optical shutter, a blocking plate, an absorber, a diffuser, and/or an optical element, and wherein the physical element is further configured to deflect and/or defocus the collected solar energy.

21. The method of claim 19, further comprising: preventing the solar power collection system from forming the solar beam using the physical element with a response time that is shorter than the time for thermal damage to occur in spacecraft structures.

22. The method of claim 19, further comprising: receiving the collected solar energy and radiate thermal energy using the physical element in order to prevent moisture or ice buildup on other portions of the solar power collection system.

23. The method of claim 18, further comprising: collecting and focusing the solar energy using a lenticular structure having a curved reflective surface into the solar beam, and redirecting the solar beam using at least one reflecting element.

24. The method of claim 23, wherein: the one or more sensors comprise one or more cameras, and the one or more cameras each have a field of view directed at one of the at least one reflecting element.

25. The method of claim 24, further comprising: receiving, at one or more processors, image data from the one or more cameras; detecting, at the one or more processors, a location of a diffuse glow with respect to the one of the reflective elements; and determining, using the one or more processors, whether the solar beam is misaligned based on the detected location of the diffuse glow.
